

Global Markets Daily: The Energy Shock Impact on Foreign Official Treasury Demand

- The US-Iran conflict has reinvigorated focus on foreign official demand for US Treasuries, as recent data suggest that foreign official investors have been net sellers of USTs since the start of the conflict. Given the context of Dollar strength and weaker external balances in Asia, we think the outflows are primarily the result of managing currency depreciation and mitigating capital outflows, rather than a structural shift.
- We find that the Dollar is one of the most meaningful drivers of valuation-adjusted foreign official Treasury demand. While the Dollar appears to have become a less significant factor in explaining foreign official demand over time, recent foreign official flows look like a return to more normal patterns.
- Despite net selling by foreign official accounts, Treasury swap spreads do not show clear evidence of sustained pressure in aggregate. Were a more persistent negative demand shift driving broader UST price action, we think it would have had a more durable impact on spreads and other measures of Treasury convenience.
- We do not think that the patterns in recent months point to a structural shift in foreign official Treasury demand. First, the overall picture of foreign official flows has been largely consistent with what we would expect based on market moves alone. Second, central banks' defense of currency arrangements typically signals a strong intention to remain tied to the Dollar and maintain ownership of USTs in the long run. And third, Treasury ownership remains supported by substantial capital market depth.
- Assuming the conflict ends, we would expect a return to the Dollar weakening trend to provide a modest tailwind to foreign official institutions' UST demand, as it had started to do once again pre-conflict.

Isabella Rosenberg
+1(212)357-7628 |
isabella.rosenberg@gs.com
Goldman Sachs & Co. LLC

The Energy Shock Impact on Foreign Official Treasury Demand

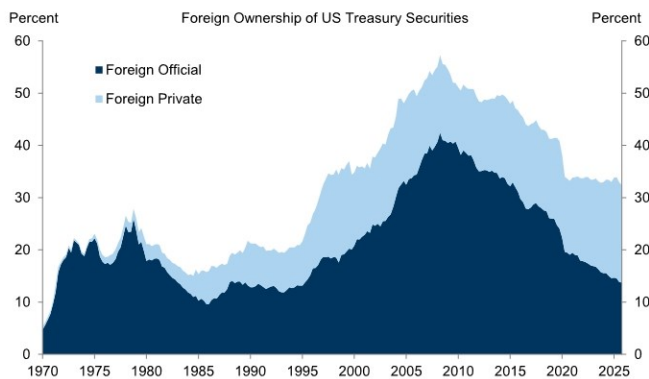
The US-Iran conflict has reinvigorated focus on foreign official demand for US Treasuries. While there is clear evidence that foreign official accounts have been net sellers of USTs since the start of the conflict, we think this is primarily due to FX intervention and mitigating capital outflows rather than a structural shift in demand patterns.

Reviewing the evolution of foreign UST demand

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

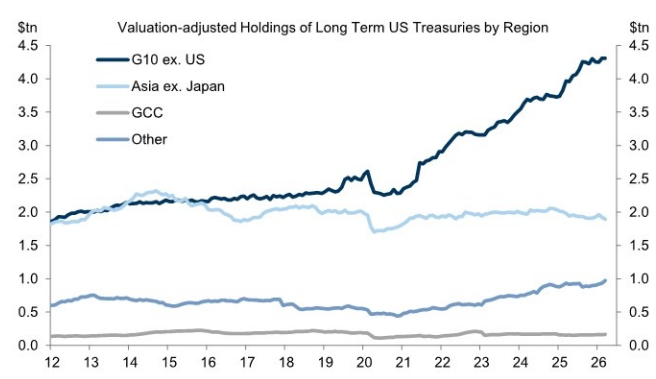
Foreign official institutions primarily manage their reserve portfolios based on policy objectives, with FX intervention and buffering capital flows being two major use cases. There are numerous reasons for large foreign official holdings of US Treasuries.¹ Among those proposed by [Bernanke \(2005\)](#) are the Asian financial crisis and other EM debt crises of the late-1990s, which forced many countries to shift from being net importers of financial capital to net exporters. Another contributing factor was the shift toward current account surpluses across many of the large oil exporting economies following the oil rallies of the late-1990s and early-to-mid-2000s. Foreign investor participation has evolved since the GFC, with foreign private investors playing a larger role as foreign official holdings have stagnated while the size of the overall market has steadily grown ([Exhibit 1](#)). This observed shift may overstate the reality due to [custodial bias](#), however, as reserve managers that use private custodians will see their holdings classified as private rather than official investment in the Treasury's TIC report.

Exhibit 1: Foreign official holdings of Treasuries have declined as a share of the market



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 2: The GCC economies have historically been a smaller, but reliable source of foreign official Treasury demand



Source: Haver Analytics, Goldman Sachs Global Investment Research

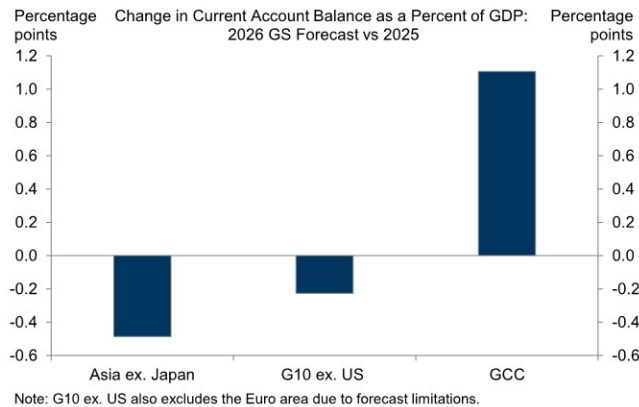
Large Gulf energy exporters have historically been a smaller, but reliable source of foreign official Treasury demand because the oil trade is overwhelmingly denominated in US Dollars ([Exhibit 2](#)), resulting in recycling of Dollar revenues from current account surpluses into Treasuries. Consequently, the halt to oil flows has increased concerns that lower USD revenues across the Gulf Cooperation Council, or GCC countries (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates) will disrupt a key source of Treasury demand. The majority of foreign Treasury ownership is allocated across G10 and Asia. However, we would highlight that some of the accumulation in G10 Treasury holdings reflects reserve assets held at large European custodians, though the ultimate owner may be based outside of Europe.

While the GCC countries have seen a [hit to growth](#) from lower production, the immediate aftermath of the oil price shock may have been slightly net positive for their aggregate [current account balance](#), as the higher price may have been able to outweigh lower export volumes ([Exhibit 3](#)). Consistent with initially more insulated external balances, the latest TIC report shows the GCC economies net purchased Treasuries in March ([Exhibit 4](#)). However, as our economists have [noted](#), the more prolonged the disruption to oil flows, the more important volume effects become relative to price

¹ This discussion is distinct from that of reserve currency choice, which is driven by many different factors, explored in greater detail [here](#).

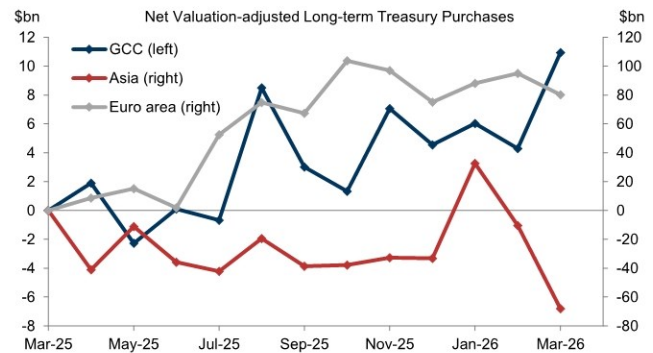
effects, which could weigh on this initial pattern over time. Some GCC economies' external balances will likely be more negatively affected than others as well, given that some countries have found alternative routes for continued oil exports. In direct contrast, for much of Asia and some of G10, paying up for oil has corresponded to a current account deterioration that has incurred net Treasury selling across these regions ([Exhibit 4](#)).

Exhibit 3: The oil shock has likely been net positive for GCC external balances and net negative for Asia



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 4: Regional Treasury demand has been consistent with current account pressures



Source: Haver Analytics, Goldman Sachs Global Investment Research

Early evidence on the Treasury demand response to the shock

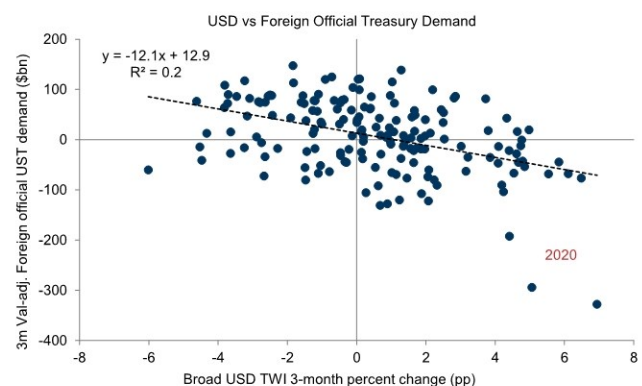
Recent data from the US Treasury and Fed custody holdings suggest that foreign official investors have net sold Treasuries since the start of the conflict as the Dollar has strengthened ([Exhibit 5](#)). Consistent with policy motivations—such as managing currency moves, mitigating capital outflows, or simply a desired mix of reserve currencies—driving valuation-adjusted foreign official flows, we find that broad USD performance is a key explanatory factor for foreign official demand ([Exhibit 6](#)). We suspect the estimated relationship between currency moves and Treasury purchases is likely biased smaller than reality, given the circular relationship between intervention and Dollar performance.

Exhibit 5: Foreign official investors net sold Treasuries at the start of the conflict as the Dollar strengthened



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 6: USD performance has historically been a key explanatory factor for foreign official demand, reflecting policy motivations



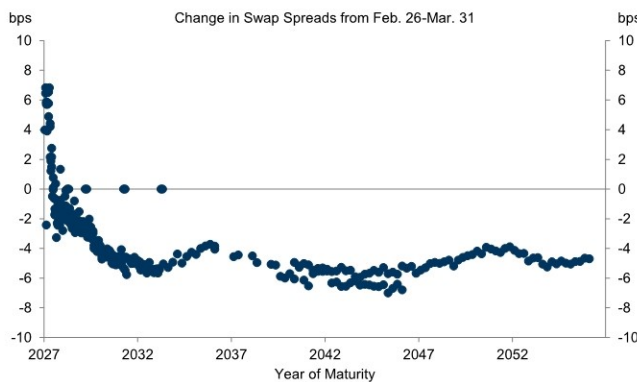
Source: Haver Analytics, Goldman Sachs Global Investment Research

A return towards more normal demand patterns rather than a structural shift

Empirically, the Dollar has become a less significant explanatory factor for foreign official demand in recent years, especially over the last year. One recent driver could be the volatility in the Dollar and yields since April 2025—our FX strategists have found that excess volatility can explain lower reserve shares—and over the longer run, reserve diversification could be contributing. However, year-to-date patterns suggest a return towards a more historically normal relationship between the Dollar and foreign official demand, as early year Dollar weakness corresponded to foreign official buying, before March's Dollar strength gave way to net selling.

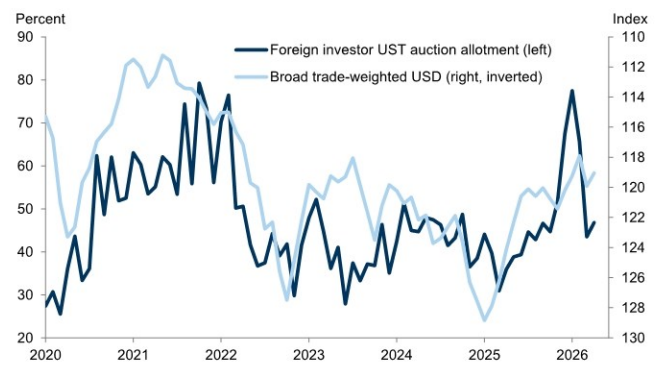
Since March, Fed custody holdings have continued to decline. While USD performance has become more differentiated, we think the pattern remains consistent with pockets of more stringent currency management. Despite the observed outflows, the move in swap spreads along the curve was relatively uniform along much of the curve beyond the very front-end (which outperformed) through March (Exhibit 7). Amid the March swap spread moves, there was relatively limited dispersion along the curve and spreads have since more than unwound the move despite some evidence that flows may have continued. And while our convenience yield framework showed a decline in Treasury convenience during the initial phase of the shock in March, it has since stabilized. We view that combination as consistent with relatively digestible selling pressure that has left limited lasting impact on the broader market.

Exhibit 7: Swap spreads have been consistent with relatively digestible selling pressure



Source: Treasury, Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Exhibit 8: A weaker Dollar supported foreign participation in Treasury auctions for much of 2025



Source: Haver Analytics, Goldman Sachs Global Investment Research

More broadly, we are skeptical that the response to the oil shock reflects a structural shift in foreign official Treasury demand for a few reasons. First, the overall picture of foreign official flows has been largely in the realm of what we would have historically expected based on market moves alone. Second, FX intervention in a managed currency is typically a sign that policymakers intend to keep it tied to the Dollar. The worse signal for structural UST demand would be evidence that some economies with managed exchange rates were abandoning the Dollar as a base currency, as this would more structurally reduce their need for USD reserves. And third, Treasury ownership as a reserve asset remains supported by substantial capital market depth, while alternatives are still somewhat constrained by the lack of available portfolio assets. For example, our FX strategists have previously shown that denominating even a small portion of GCC oil exports in RMB would lead to a significant increase in RMB reserves, to the extent that it would be both difficult to invest and quickly erode the reserve coverage necessary to protect their current FX management system.

USD strength following the conflict interrupted a key Dollar weakening trend that had supported stronger foreign investor participation in Treasury auctions through 2025 and broader demand ([Exhibit 8](#)). Assuming an end to the conflict, we would expect a return to the Dollar weakening trend and lower volatility to be a modest tailwind to foreign official UST demand.

TRADE IDEAS



Best Trade Ideas Across Assets

For pricing, charts, and a list of previous recommendations, please visit our [Trade Ideas page](#).

1. Stay short SGD/MYR, opened January 24, 2026, at 3.13, with a target at 2.90 and a stop at 3.30, currently trading at 3.11.
2. Stay long TRY, NGN and KZT against the USD, as an equally weighted basket, opened February 18, 2026, at 0%, with a total return target of 7.5%, and a revised stop of 0%, currently trading at 1.9%.
3. Stay long USD/SEK, opened March 20, 2026, at 9.3443, with a target at 9.65 and a stop at 9.00, currently trading at 9.3089.
4. Stay long 3y France, Spain, Italy vs OIS (equally weighted), opened April 10, 2026, at 0.33, with a target at 0.23 and a revised stop at 0.32, currently trading at 0.30.
5. Stay long MSCI Korea and Taiwan (equal weight) vs. short MSCI India, Philippines, Thailand (equal weight), opened April 16, 2026, at 100, with a revised target of 135 and a revised stop of 115, currently trading at 130.
6. Stay short EUR/HUF, opened 17 April 2026, at 361, with a target of 350 and a stop of 372, currently trading at 356.
7. Stay long 3y SOFR swap spread, opened 17 April 2026, at -22.6bp, with a target at -18bp inclusive of carry; and a revised stop at -23bp, currently trading at -20.5bp.
8. 2s5s CAD steepener, opened 24 April 2026, at 20bp, with a target at 35bp and a revised stop at 20bp, current trading at 23bp.
9. Stay short USD/EGP, opened 24 April 2026, at 0%, with a total return target at 6% and a stop at -3%, currently trading at 1.8%.
10. Stay long 5y5y EUR OIS – HICP, opened 01 May 2026, at 1.09%, with a target at 0.88% and a stop at 1.22%, currently trading at 1.03%.
11. Stay long 5y Receivers on 3m 2s5s10s Receiver Fly (bp running), opened 09 May 2026, at 1bp, with a target at 10bp and a stop at -5bp, currently trading at -1bp.

Global Interest Rates Strategy

George Cole
+44(20)7552-1214
george.cole@gs.com
Goldman Sachs International

William Marshall
+1(212)357-0413
william.c.marshall@gs.com
Goldman Sachs & Co. LLC

Simon Freyenet
+44(20)7774-5017
simon.freyenet@gs.com
Goldman Sachs Bank Europe SE - Paris
Branch

Isabella Rosenberg
+1(212)357-7628
isabella.rosenberg@gs.com
Goldman Sachs & Co. LLC

Friedrich Schaper
+1(917)343-3214
friedrich.schaper@gs.com
Goldman Sachs & Co. LLC

Loic Mathys
+44(20)7051-1664
loic.mathys@gs.com
Goldman Sachs International

Disclosure Appendix

Reg AC

I, Isabella Rosenberg, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Isabella Rosenberg Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.